

Agentic Malware Analysis: From Task Automation to Deep Analysis

Binary Cartography

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About Tim

- independent security researcher, consultant & trainer
- PhD in binary program analysis & reverse engineering
- focus on (de)obfuscation & automation
- trainings on reverse engineering & software deobfuscation





What is agentic malware analysis?



String decryption, API resolving, multi-stage



From task automation to analysis pipelines

Quick Recap

- agent = LLM + tools + loop + goal
- plans, calls tools autonomously, iterates on results
- sandboxed Docker environment with all tools pre-installed
- Claude Code / Codex as agent runtime

- MCP: standardized protocol for agent-to-tool calls
- headless MCP servers for Ghidra and Binary Ninja
- skills: reusable workflows that guide agent behavior

Agentic Malware Analysis

Why Agents for Malware Analysis?

- malware analysis is full of repetitive, structured tasks
- with guidance, agents can support analysis well
- e.g. string decoding, API resolving, unpacking, triage
- beyond that: autonomous analysis in structured pipelines

Task Automation

- analyst identifies a task, agent executes it autonomously
- decode strings, resolve imports, extract configs, triage
- agent iterates: reads code, writes scripts, validates results
- analyst stays in control, agent handles the tedious parts

String Decryption

- malware encrypts strings to hide IoCs, C2s, commands
- decryption routine called at runtime for each string
- agent: find routine, understand logic, decrypt all strings

API Resolving

- malware avoids static imports to evade detection
- resolves API functions at runtime via hash comparison
- agent: analyze code, write brute forcer, annotate database

<https://blog.talosintelligence.com/qilin-edr-killer/>

Multi-Stage Analysis

Multi-Stage Samples

- malware decrypts and loads additional modules at runtime
- each stage may use different protection techniques
- agent: unpack stages, extract configs and artifacts

[https://www.amnesty.org/en/latest/research/2020/09/
german-made-finspy-spyware-found-in-egypt-and-mac-and-linux-versions-revealed/](https://www.amnesty.org/en/latest/research/2020/09/german-made-finspy-spyware-found-in-egypt-and-mac-and-linux-versions-revealed/)

From Tasks to Pipelines

- so far: agents supporting individual analysis tasks
- but: can also orchestrate full analysis workflows
- structured, multi-phase investigation
- enforce thorough, structured documentation

Why Pipelines Work

- broad tool-driven fingerprinting as first phase
- agent reviews artifacts, sets own priorities
- builds hypotheses, validates through deep analysis
- persistent state — resumable across sessions

https://synthesis.to/2026/03/18/agentic_malware_analysis.html

Wrap-Up

- results are non-deterministic
- context window limits depth of single-pass analysis
- complex tasks still need analyst guidance and steering
- verification of agent output remains essential

- agents handle focused malware RE tasks reliably
- pipelines enable deeper, autonomous investigation
- analyst expertise decides where and how to steer
- skills and workflows can abstract away many layers

Summary

- task automation: string decryption, API resolving, unpacking
- deep analysis: structured pipelines with autonomous agents
- from focused tasks to end-to-end malware investigation

<https://github.com/mrphrazer/binary-cartography>

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